

Exp No: 27

BINARY SEMAPHORE IN SYSTEMC

Program

```
#include <systemc.h>

SC_MODULE(binary_semaphore) {
    int sem;
    sc_event ev;
    void task1() {
        while(true) {
            wait(1, SC_SEC);
            if(sem == 1) {
                sem = 0;
                cout << "Task1 Entered Critical Section at " << sc_time_stamp() << endl;
                wait(2, SC_SEC);
                sem = 1;
                ev.notify();
            }
            else {
                wait(ev);
            }
        }
    }

    void task2() {
        while(true) {
            wait(2, SC_SEC);

            if(sem == 1) {
                sem = 0;
                cout << "Task2 Entered Critical Section at " << sc_time_stamp() << endl;
                wait(1, SC_SEC);
                sem = 1;
            }
        }
    }
}
```

```

        ev.notify();
    }
    else {
        wait(ev);
    }
}
}
}

```

```

SC_CTOR(binary_semaphore) {
    sem = 1;
    SC_THREAD(task1);
    SC_THREAD(task2);
}

};

int sc_main(int argc, char* argv[]) {
    binary_semaphore obj("Binary_Semaphore");
    sc_start(10, SC_SEC);
    return 0;
}

```

Output

Compile done. Starting run...

```

SystemC 2.3.3-Accellera --- May 23 2020 14:33:56
Copyright (c) 1996-2018 by all Contributors,
ALL RIGHTS RESERVED
Task1 Entered Critical Section at 1 s
Task1 Entered Critical Section at 4 s
Task1 Entered Critical Section at 7 s

```